

Who Bears the Burden of Bank Capital Regulation?

Buffers, Risk Weights, and the Supply of Credit

Juan Zurita

School of Economics, University of Edinburgh

October 1, 2026

Motivation

Why requirements exist: equity absorbs unexpected losses, so banks stay solvent in downturns—protecting insured deposits and financial stability.

The debate over how far they should rise has been conducted in aggregates:
stability benefits of a higher requirement vs. the output cost of getting there

Why requirements exist: equity absorbs unexpected losses, so banks stay solvent in downturns—protecting insured deposits and financial stability.

The debate over how far they should rise has been conducted in aggregates:
stability benefits of a higher requirement vs. the output cost of getting there

But the regime is not an aggregate lever—it is deliberately differentiated:

- Risk weights: C&I loans $\omega_B = 1.00$ vs. mortgages $\omega_M = 0.50$
- Size tiers: surcharges at \$10B and \$50B asset thresholds (post-GFC)

Why requirements exist: equity absorbs unexpected losses, so banks stay solvent in downturns—protecting insured deposits and financial stability.

The debate over how far they should rise has been conducted in aggregates:
stability benefits of a higher requirement vs. the output cost of getting there

But the regime is not an aggregate lever—it is deliberately differentiated:

- Risk weights: C&I loans $\omega_B = 1.00$ vs. mortgages $\omega_M = 0.50$
 - Size tiers: surcharges at \$10B and \$50B asset thresholds (post-GFC)
- Both margins presume regulators know *where a tightening binds*.

Who bears the burden of a capital requirement tightening?

- ▶ Which **sectors** absorb the contraction in credit?
- ▶ Which **banks** transmit it?
- ▶ What is the **aggregate cost**—and does design matter for it?

Paper in a Nutshell

Two facts (Call Reports, 4,800 banks, 1994–2019; identified shocks):

- ▶ **Across sectors:** C&I lending contracts $\sim$ a third more than real estate—the risk-weight wedge

Paper in a Nutshell

Two facts (Call Reports, 4,800 banks, 1994–2019; identified shocks):

- ▶ **Across sectors:** C&I lending contracts $\sim$ a third more than real estate—the risk-weight wedge
- ▶ **Across banks:** banks with the least room above *their own* requirement cut $1.32\times$ more than those with the most (90% CI [1.01, 1.62])—and it is that room, not size

Paper in a Nutshell

Two facts (Call Reports, 4,800 banks, 1994–2019; identified shocks):

- ▶ **Across sectors:** C&I lending contracts $\sim$ a third more than real estate—the risk-weight wedge
- ▶ **Across banks:** banks with the least room above *their own* requirement cut $1.32\times$ more than those with the most (90% CI [1.01, 1.62])—and it is that room, not size

A model disciplined by those facts:

- ▶ Reproduces the gradient at $1.23\times$, **untargeted** (though not the hump—simulated responses deepen without reverting)
- ▶ A +1pp tightening costs 1.4–2.4% of credit-market surplus

Paper in a Nutshell

Two facts (Call Reports, 4,800 banks, 1994–2019; identified shocks):

- ▶ **Across sectors:** C&I lending contracts $\sim$ a third more than real estate—the risk-weight wedge
- ▶ **Across banks:** banks with the least room above *their own* requirement cut $1.32\times$ more than those with the most (90% CI [1.01, 1.62])—and it is that room, not size

A model disciplined by those facts:

- ▶ Reproduces the gradient at $1.23\times$, **untargeted** (though not the hump—simulated responses deepen without reverting)
- ▶ A +1pp tightening costs 1.4–2.4% of credit-market surplus

The burden is organised by *distance to the frontier*, and the statutory margin explains just 1.8% of its variance.

Level of requirements $\Rightarrow$ aggregate cost. *Design* $\Rightarrow$ **incidence**.

What: how much capital each bank holds *above its own requirement*

- Call that gap the *voluntary buffer* $D_{i,t}$ — heterogeneous funding frictions γ_i generate it
- **This paper carries the whole distribution of $D_{i,t}$ across banks as a state variable**

Contribution

What: how much capital each bank holds *above its own requirement*

- Call that gap the *voluntary buffer* $D_{i,t}$ — heterogeneous funding frictions γ_i generate it
- **This paper carries the whole distribution of $D_{i,t}$ across banks as a state variable**

Context: the closest quantitative models do not carry that distribution

- **Elenev, Landvoigt, and Van Nieuwerburgh (2021)** — one intermediary; failure risk i.i.d., survivors recapitalised each period $\Rightarrow$ incidence is *sector-level* by construction
- **Corbae and Erasmo (2021)** — two bank types; buffers disciplined at *two group means*, validated on a *size-sorted* regression

Source: Consolidated Reports of Condition and Income (Call Reports, FDIC)

- ▶ ~4,800 U.S. commercial banks, quarterly, 1994–2019
- ▶ Loans decomposed: real estate, C&I, consumption, credit card
- ▶ Macro controls: real GDP growth, federal funds rate (FRED)

¹Economic Growth, Regulatory Relief, and Consumer Protection Act (2018), which raised the automatic enhanced-prudential-standards threshold from \$50B to \$250B. The \$50B tier governs almost the whole 1994–2019 sample.

Source: Consolidated Reports of Condition and Income (Call Reports, FDIC)

- ▶ ~4,800 U.S. commercial banks, quarterly, 1994–2019
- ▶ Loans decomposed: real estate, C&I, consumption, credit card
- ▶ Macro controls: real GDP growth, federal funds rate (FRED)

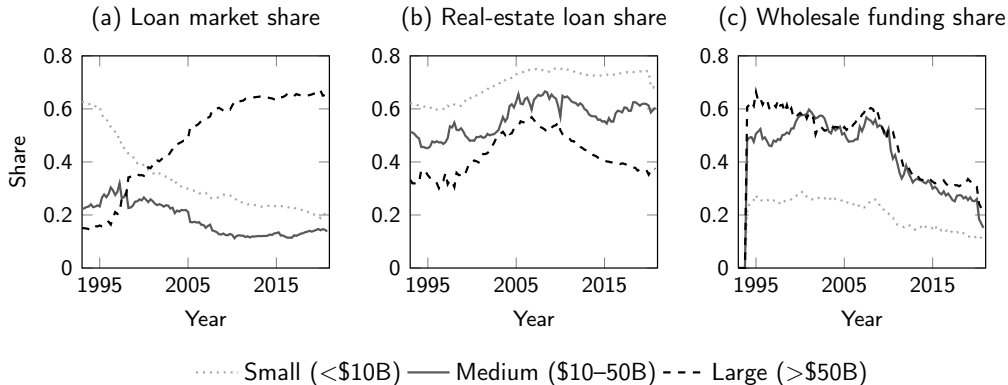
Size cohorts (Dodd-Frank/EGRRCPA¹ thresholds):

- ▶ Community, $< \$10\text{B}$ — baseline κ
- ▶ Regional, $\$10\text{--}50\text{B}$ — $\kappa + 1.5\text{pp}$ (Dodd-Frank)
- ▶ SIFI, $> \$50\text{B}$ — $\kappa + 2.5\text{pp}$ (G-SIB surcharge)

Regulatory wedge

¹Economic Growth, Regulatory Relief, and Consumer Protection Act (2018), which raised the automatic enhanced-prudential-standards threshold from \$50B to \$250B. The \$50B tier governs almost the whole 1994–2019 sample.

Market Concentration and Asset Specialisation



Funding Capacity and Regulatory Slack

Banks that fund themselves at the wholesale margin self-insure.

Those with stable deposit franchises hold *thinner* buffers.

Funding Capacity and Regulatory Slack

Banks that fund themselves at the wholesale margin self-insure.

Those with stable deposit franchises hold *thinner* buffers.

Regulatory Slack $D_{i,t} \equiv r_{i,t} - \kappa_{i,t}$ (distance to constraint, pp)

$r_{i,t}$ is the bank's total risk-based capital ratio (regulatory capital / RWA, Call Reports)

$\kappa_{i,t}$ is bank i 's own statutory requirement—baseline, +1.5pp (DFA) or +2.5pp (G-SIB)

Funding Capacity $\equiv (\text{core retail} + \text{NIB deposits}) / \text{total liabilities}$

Funding Capacity and Regulatory Slack

Banks that fund themselves at the wholesale margin self-insure.

Those with stable deposit franchises hold *thinner* buffers.

Regulatory Slack $D_{i,t} \equiv r_{i,t} - \kappa_{i,t}$ (distance to constraint, pp)

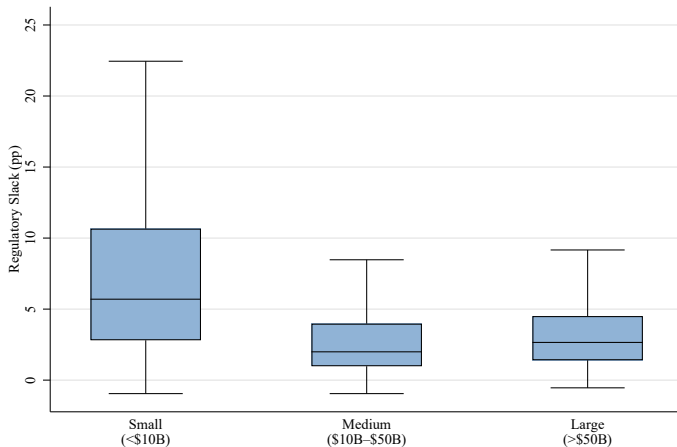
$r_{i,t}$ is the bank's total risk-based capital ratio (regulatory capital / RWA, Call Reports)

$\kappa_{i,t}$ is bank i 's own statutory requirement—baseline, +1.5pp (DFA) or +2.5pp (G-SIB)

Funding Capacity $\equiv$ (core retail + NIB deposits) / total liabilities

- ▶ A one-standard-deviation higher core-deposit share (≈ 0.10) goes with **2.0–2.6pp** less regulatory slack — about 40% of the median buffer of 5.7pp
- ▶ Strong comparing *different* banks, near-zero for a *given* bank over time — a funding model is structural, not a quarterly choice

Buffer Heterogeneity Cuts Across Size Tiers



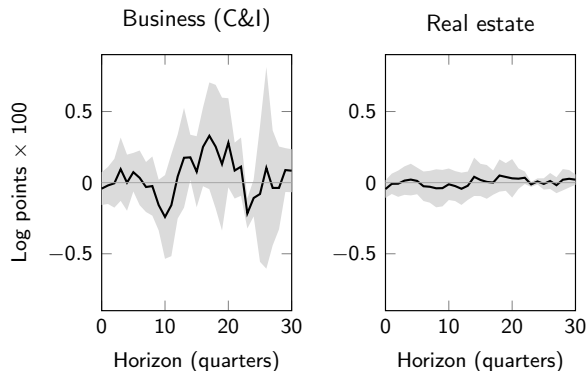
Source: US Call Reports, 1993Q2–2019Q4.
Regulatory slack: equity-to-RWA ratio minus effective requirement (κ_{eff}), winsorized at 1/99.

Fact 1: Sectoral Asymmetry in Credit Responses

Identification: Narratively identified capital-requirement shocks (Drechsel and Miura, 2025);
within-bank local projections

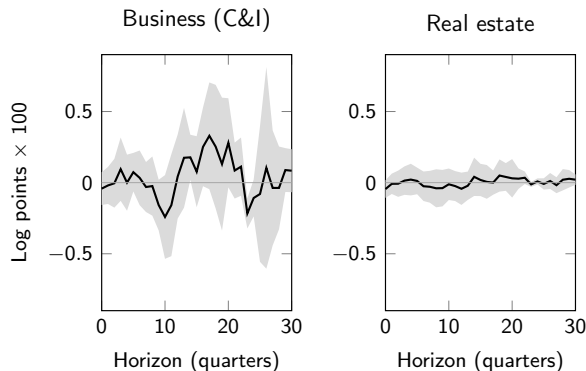
Fact 1: Sectoral Asymmetry in Credit Responses

Identification: Narratively identified capital-requirement shocks (Drechsel and Miura, 2025); within-bank local projections



Fact 1: Sectoral Asymmetry in Credit Responses

Identification: Narratively identified capital-requirement shocks (Drechsel and Miura, 2025); within-bank local projections

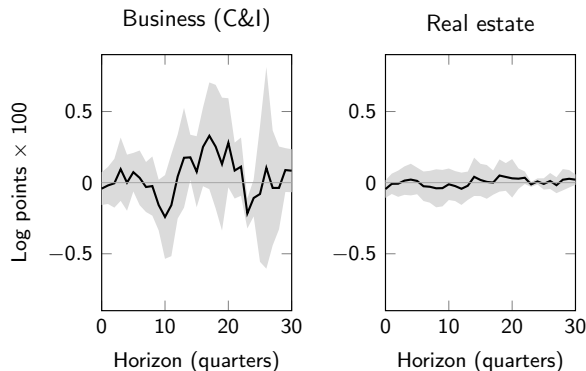


Asset Substitution Margin

When the constraint binds, banks shed high- ω assets first: a dollar of C&I frees the most required capital.

Fact 1: Sectoral Asymmetry in Credit Responses

Identification: Narratively identified capital-requirement shocks (Drechsel and Miura, 2025); within-bank local projections



Asset Substitution Margin

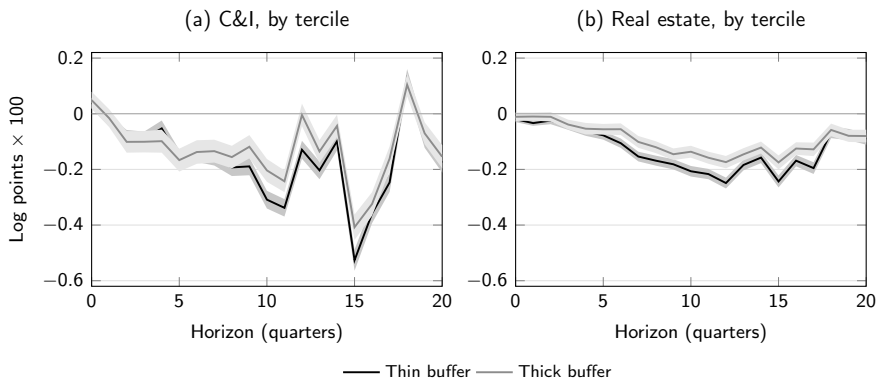
When the constraint binds, banks shed high- ω assets first: a dollar of C&I frees the most required capital.

Scale Deleveraging Multiplier

$$\frac{\partial RWA}{\partial A^B} = \omega_B > \omega_M = \frac{\partial RWA}{\partial A^M}$$

Each dollar of C&I cut yields twice the regulatory relief of a mortgage cut.

Fact 2: The Buffer Gradient



Banks sorted by lagged regulatory slack $D_{i,t-1}$.

Thin-buffer banks contract $1.32\times$ more in C&I; the model reproduces $1.23\times$, **untargeted**.

Details

Precision

What the Buffer Gradient Identifies

A capital constraint alone cannot produce it.

- ▶ With regulation entering only as a constraint, an unconstrained bank's first-order condition contains no κ : $\partial A_i^j / \partial \kappa = 0$ *exactly*

What the Buffer Gradient Identifies

A capital constraint alone cannot produce it.

- ▶ With regulation entering only as a constraint, an unconstrained bank's first-order condition contains no κ : $\partial A_i^j / \partial \kappa = 0$ *exactly*
- ▶ Fewer than 3% of bank-quarters sit at or inside the frontier

What the Buffer Gradient Identifies

A capital constraint alone cannot produce it.

- ▶ With regulation entering only as a constraint, an unconstrained bank's first-order condition contains no κ : $\partial A_i^j / \partial \kappa = 0$ *exactly*
- ▶ Fewer than 3% of bank-quarters sit at or inside the frontier
- ▶ Yet the gradient spans the whole distribution — the *thick* tercile contracts -0.13% , significantly

What the Buffer Gradient Identifies

A capital constraint alone cannot produce it.

- ▶ With regulation entering only as a constraint, an unconstrained bank's first-order condition contains no κ : $\partial A_i^j / \partial \kappa = 0$ *exactly*
- ▶ Fewer than 3% of bank-quarters sit at or inside the frontier
- ▶ Yet the gradient spans the whole distribution — the *thick* tercile contracts -0.13% , significantly

So the gradient identifies a cost of capital that is not risk-weighted:

$$\phi_{i,j}(\kappa) = (c + z \omega_j) \max(\kappa - \bar{\kappa}_i, 0)$$

What the Buffer Gradient Identifies

A capital constraint alone cannot produce it.

- ▶ With regulation entering only as a constraint, an unconstrained bank's first-order condition contains no κ : $\partial A_i^j / \partial \kappa = 0$ *exactly*
- ▶ Fewer than 3% of bank-quarters sit at or inside the frontier
- ▶ Yet the gradient spans the whole distribution — the *thick* tercile contracts -0.13% , significantly

So the gradient identifies a cost of capital that is not risk-weighted:

$$\phi_{i,j}(\kappa) = (c + z \omega_j) \max(\kappa - \bar{\kappa}_i, 0)$$

- ▶ $c = 0.148$ (not risk-weighted), $z = 0.021$ (risk-weighted)
set on the two sectoral aggregates, both buffer gradients **untargeted**

Model

Facts recap

Model: Environment

Dynamic, partial equilibrium, heterogeneous-agent.

- ▶ Continuum of infinitely lived, risk-neutral banks; equity state $e_{i,t}$
- ▶ Two sectors—real estate ($j = M$) and C&I ($j = B$)—with inverse loan demand $r_{j,t} = \theta_j - \eta_j L_t^j$

Model: Environment

Dynamic, partial equilibrium, heterogeneous-agent.

- ▶ Continuum of infinitely lived, risk-neutral banks; equity state $e_{i,t}$
- ▶ Two sectors—real estate ($j = M$) and C&I ($j = B$)—with inverse loan demand $r_{j,t} = \theta_j - \eta_j L_t^j$
- ▶ Core deposits $d_{i,C} \leq \bar{d}_{i,C}$, plus wholesale $d_{i,t}^W$ at quadratic cost $\frac{\gamma_i}{2} (d_{i,t}^W)^2$
- ▶ $\gamma_{\text{small}} \gg \gamma_{\text{large}} \rightarrow 0$: small banks face steep wholesale costs — this is where buffer heterogeneity comes from

Model: Environment

Dynamic, partial equilibrium, heterogeneous-agent.

- ▶ Continuum of infinitely lived, risk-neutral banks; equity state $e_{i,t}$
- ▶ Two sectors—real estate ($j = M$) and C&I ($j = B$)—with inverse loan demand $r_{j,t} = \theta_j - \eta_j L_t^j$
- ▶ Core deposits $d_{i,C} \leq \bar{d}_{i,C}$, plus wholesale $d_{i,t}^W$ at quadratic cost $\frac{\gamma_i}{2} (d_{i,t}^W)^2$
- ▶ $\gamma_{\text{small}} \gg \gamma_{\text{large}} \rightarrow 0$: small banks face steep wholesale costs — this is where buffer heterogeneity comes from
- ▶ One aggregate disturbance: κ_t follows an AR(1); bank i faces $\kappa_{i,t} = \kappa_t +$ its statutory tier add-on
- ▶ Two occasionally binding constraints: regulatory capital, and dividend non-negativity

The Dynamic Bank Problem (I): The Problem

Banks choose lending and dividends to maximise the discounted dividend stream:

$$V(e_{i,t}, A_{i,t-1}^M, A_{i,t-1}^B; \Omega_t) = \max \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^{\tau} c_{i,t+\tau}$$

Ω_t is the aggregate state: the current requirement and the two sectoral credit aggregates that clear the loan markets.

The Dynamic Bank Problem (I): The Problem

Banks choose lending and dividends to maximise the discounted dividend stream:

$$V(e_{i,t}, A_{i,t-1}^M, A_{i,t-1}^B; \Omega_t) = \max \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^\tau c_{i,t+\tau}$$

Ω_t is the aggregate state: the current requirement and the two sectoral credit aggregates that clear the loan markets.

Subject to the balance-sheet identity — wholesale funding is *residual*, not chosen:

$$d_{i,t}^W = A_{i,t}^M + A_{i,t}^B - e_{i,t} - \bar{d}_{i,C}$$

and to the law of motion for equity, which carries returns, costs and the levy into $e_{i,t+1}$.

Equity law of motion

The Dynamic Bank Problem (I): The Problem

Banks choose lending and dividends to maximise the discounted dividend stream:

$$V(e_{i,t}, A_{i,t-1}^M, A_{i,t-1}^B; \Omega_t) = \max \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^{\tau} c_{i,t+\tau}$$

Ω_t is the aggregate state: the current requirement and the two sectoral credit aggregates that clear the loan markets.

Subject to the balance-sheet identity — wholesale funding is *residual*, not chosen:

$$d_{i,t}^W = A_{i,t}^M + A_{i,t}^B - e_{i,t} - \bar{d}_{i,C}$$

and to the law of motion for equity, which carries returns, costs and the levy into $e_{i,t+1}$.

Equity law of motion

and to two occasionally binding constraints:

$$\underbrace{\omega_M A_{i,t}^M + \omega_B A_{i,t}^B}_{\text{regulatory capital}} \leq \frac{e_{i,t}}{\kappa_{i,t}}, \quad \underbrace{c_{i,t}}_{\text{no equity issuance}} \geq 0$$

The Dynamic Bank Problem (II): The Euler Equation

Optimal lending in sector j satisfies:

$$\beta \mathbb{E}_t \left[(1 + \mu_{i,t+1}) \left(r_{j,t} - \varepsilon_{i,t+1}^j - \psi_j A_{i,t}^j - r_{W,t} - \gamma_i d_{i,t}^W - \phi_{i,j}(\kappa_{i,t}) \right) \right] = \lambda_{i,t} \omega_j$$

The Dynamic Bank Problem (II): The Euler Equation

Optimal lending in sector j satisfies:

$$\beta \mathbb{E}_t \left[(1 + \mu_{i,t+1}) \left(r_{j,t} - \varepsilon_{i,t+1}^j - \psi_j A_{i,t}^j - r_{W,t} - \gamma_i d_{i,t}^W - \phi_{i,j}(\kappa_{i,t}) \right) \right] = \lambda_{i,t} \omega_j$$

- ▶ $\lambda_{i,t}$: shadow cost of the capital constraint — zero when the bank is away from the frontier
- ▶ $\mu_{i,t}$: multiplier on $c_{i,t} \geq 0$ — internal funds are worth more than one when the bank would cut dividends below zero and cannot

The Dynamic Bank Problem (II): The Euler Equation

Optimal lending in sector j satisfies:

$$\beta \mathbb{E}_t \left[(1 + \mu_{i,t+1}) \left(r_{j,t} - \varepsilon_{i,t+1}^j - \psi_j A_{i,t}^j - r_{W,t} - \gamma_i d_{i,t}^W - \phi_{i,j}(\kappa_{i,t}) \right) \right] = \lambda_{i,t} \omega_j$$

- ▶ $\lambda_{i,t}$: shadow cost of the capital constraint — zero when the bank is away from the frontier
- ▶ $\mu_{i,t}$: multiplier on $c_{i,t} \geq 0$ — internal funds are worth more than one when the bank would cut dividends below zero and cannot
- ▶ $\phi_{i,j}$: per-unit cost of regulatory capital, paid *whether or not* the constraint binds (bank-specific via $\bar{\kappa}_i$)
- ▶ $\omega_B > \omega_M \Rightarrow$ higher regulatory cost per unit C&I $\Rightarrow$ C&I contracts more

Where these terms come from

Neural-Network Solution

Challenge: occasionally binding constraints $\Rightarrow$ kinked policy functions.

Neural-Network Solution

Challenge: occasionally binding constraints $\Rightarrow$ kinked policy functions.

Approach: parameterise policy with a multi-layer perceptron (MLP):

$$\begin{bmatrix} \hat{A}_{i,t}^M \\ \hat{A}_{i,t}^B \\ \hat{c}_{i,t} \\ \hat{\lambda}_{i,t} \\ \hat{\mu}_{i,t} \end{bmatrix} = \mathcal{A}(S_{i,t}; \Theta) \quad S_{i,t} \equiv (e_{i,t}, A_{i,t-1}^M, A_{i,t-1}^B, \Omega_t)$$

Neural-Network Solution

Challenge: occasionally binding constraints $\Rightarrow$ kinked policy functions.

Approach: parameterise policy with a multi-layer perceptron (MLP):

$$\begin{bmatrix} \hat{A}_{i,t}^M \\ \hat{A}_{i,t}^B \\ \hat{c}_{i,t} \\ \hat{\lambda}_{i,t} \\ \hat{\mu}_{i,t} \end{bmatrix} = \mathcal{A}(S_{i,t}; \Theta) \quad S_{i,t} \equiv (e_{i,t}, A_{i,t-1}^M, A_{i,t-1}^B, \Omega_t)$$

$\Omega_t \equiv (\kappa_t, L_t^M, L_t^B)$: the current requirement and the two sectoral credit aggregates.

Neural-Network Solution

Challenge: occasionally binding constraints $\Rightarrow$ kinked policy functions.

Approach: parameterise policy with a multi-layer perceptron (MLP):

$$\begin{bmatrix} \hat{A}_{i,t}^M \\ \hat{A}_{i,t}^B \\ \hat{c}_{i,t} \\ \hat{\lambda}_{i,t} \\ \hat{\mu}_{i,t} \end{bmatrix} = \mathcal{A}(S_{i,t}; \Theta) \quad S_{i,t} \equiv (e_{i,t}, A_{i,t-1}^M, A_{i,t-1}^B, \Omega_t)$$

$\Omega_t \equiv (\kappa_t, L_t^M, L_t^B)$: the current requirement and the two sectoral credit aggregates.

- **Structural loss:** squared Euler residuals + soft KT penalty
- **Market clearing:** imposed analytically via closed-form equilibrium rate
- **Architecture:** 4 hidden layers $\times$ 256 nodes, Swish activation
- **Optimizer:** AdamW with cosine annealing

Network

Method

Non-convexity

Solution accuracy

What the policy pins down

SMM Estimation Strategy

Estimate: $\theta = (\rho_{\kappa}, \sigma_{\kappa})$ — persistence and volatility of the capital-requirement process

Targets: 16 within-bank LP coefficients

C&I and RE responses at $\mathcal{H} = \{1, 2, 4, 6, 8, 12, 16, 20\}$

SMM Estimation Strategy

Estimate: $\theta = (\rho_\kappa, \sigma_\kappa)$ — persistence and volatility of the capital-requirement process

Targets: 16 within-bank LP coefficients

C&I and RE responses at $\mathcal{H} = \{1, 2, 4, 6, 8, 12, 16, 20\}$

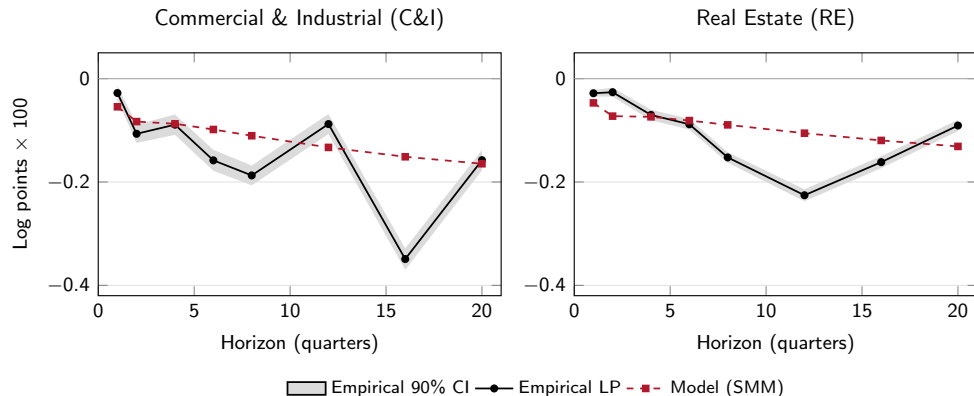
Identification, in words:

- ▶ $\rho_\kappa \leftarrow$ how slowly the IRF decays at medium horizons
- ▶ $\sigma_\kappa \leftarrow$ how large the impact response is
- ▶ Two parameters, sixteen moments — the process is over-identified.

Weighting and algorithm

Calibration

Estimation: Model vs. Data

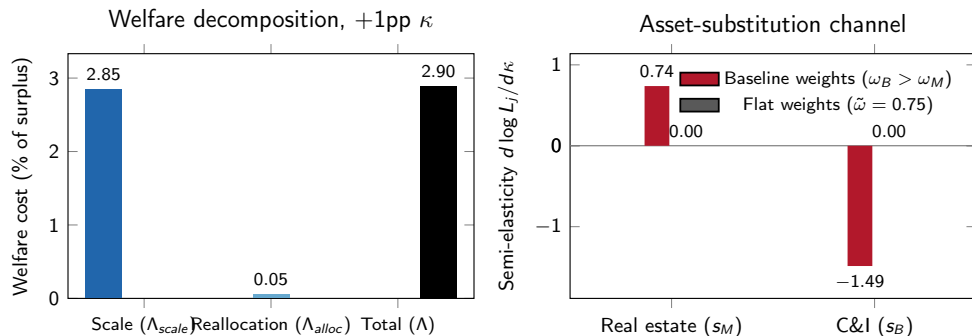


$\hat{\rho}_{\kappa} = 0.9987$ (s.e. 0.004), $\hat{\sigma}_{\kappa} = 0.00040$ (s.e. 0.00002) — interior, within 1 s.e. of a unit root.

Moment fit

Robustness

Gross Cost and Decomposition



Left: CEV decomposition. Right: constrained-regime semi-elasticities.

Counterfactuals: neither a uniform $\tilde{\kappa}$ nor flattened risk weights changes the aggregate burden.

Counterfactuals

Details

CEV framework

Magnitudes in context

Who Bears It? Incidence Across Banks

The statute sorts by size. The mechanism sorts by buffer.

- ▶ Statutory tier explains 1.8% of the cross-sectional variance in regulatory slack

Who Bears It? Incidence Across Banks

The statute sorts by size. The mechanism sorts by buffer.

- ▶ Statutory tier explains 1.8% of the cross-sectional variance in regulatory slack
- ▶ 86% of thin-buffer banks (47% by assets) face the *lowest* statutory requirement
- ▶ 84% of banks above \$50bn already hold the thinnest buffers

Who Bears It? Incidence Across Banks

The statute sorts by size. The mechanism sorts by buffer.

- ▶ Statutory tier explains 1.8% of the cross-sectional variance in regulatory slack
- ▶ 86% of thin-buffer banks (47% by assets) face the *lowest* statutory requirement
- ▶ 84% of banks above \$50bn already hold the thinnest buffers
- ▶ **Thin-buffer banks:** 32% of lending, 34% of the burden — concentration 1.06
- ▶ **Thick-buffer banks:** 35% of lending, 33% of the burden — concentration 0.94

Who Bears It? Incidence Across Banks

The statute sorts by size. The mechanism sorts by buffer.

- ▶ Statutory tier explains 1.8% of the cross-sectional variance in regulatory slack
- ▶ 86% of thin-buffer banks (47% by assets) face the *lowest* statutory requirement
- ▶ 84% of banks above \$50bn already hold the thinnest buffers
- ▶ **Thin-buffer banks:** 32% of lending, 34% of the burden — concentration 1.06
- ▶ **Thick-buffer banks:** 35% of lending, 33% of the burden — concentration 0.94

Thin-buffer banks cut 13% more *per dollar lent*, but they are smaller — so the burden is disproportionate per unit of credit, not concentrated on a small group.

Who bears the burden of bank capital regulation?

- ▶ **Which sectors:** risk weights concentrate the contraction in C&I ($1.31\times$ vs real estate)
- ▶ **Which banks:** distance to the frontier, not size ($1.32\times$ in the data, $1.23\times$ untargeted in the model)

Who bears the burden of bank capital regulation?

- ▶ **Which sectors:** risk weights concentrate the contraction in C&I ($1.31\times$ vs real estate)
- ▶ **Which banks:** distance to the frontier, not size ($1.32\times$ in the data, $1.23\times$ untargeted in the model)

The level of requirements sets the aggregate cost (1.4–2.4% of credit-market surplus per +1pp); their design moves only who pays it.

Gross transition costs excluding stability benefits.

GE channels (borrower substitution to non-banks, loan-rate adjustment) would attenuate the real-side impact; aggregate-demand feedback could amplify it.

Thank you

Who Bears the Burden of Bank Capital Regulation?

Juan Zurita | University of Edinburgh

juan.zurita@ed.ac.uk

Interpreting the Sectoral Asymmetry

Why C&I contracts more than RE:

► **Asset Substitution Margin:** when the constraint binds, banks shed high- ω assets (C&I, $\omega_B = 1$) to free the most required capital per dollar cut

► **Scale Deleveraging Multiplier:** $\partial RWA / \partial A^B = \omega_B > \omega_M = \partial RWA / \partial A^M$

Each dollar of C&I cut yields twice the regulatory relief of a mortgage cut

► **Consumer credit unresponsive:** high interest margins provide sufficient cash-flow buffers to absorb the regulatory cost without volume rationing

Result: the sectoral composition of the contraction is determined by the risk-weight schedule, not by loan demand.

SMM Moment Fit

Sector	Horizon h	$\hat{b}_h$ (data)	s.e.	b_h (model)	t
C&I	1	-0.028	0.008	-0.054	+3.20
	2	-0.107	0.011	-0.083	-2.20
	4	-0.089	0.012	-0.087	-0.14
	6	-0.158	0.012	-0.098	-4.85
	8	-0.187	0.012	-0.110	-6.57
	12	-0.088	0.012	-0.133	+3.84
	16	-0.349	0.013	-0.151	-15.73
	20	-0.158	0.012	-0.165	+0.58
RE	1	-0.028	0.004	-0.047	+4.87
	2	-0.026	0.005	-0.073	+8.80
	4	-0.070	0.006	-0.074	+0.59
	6	-0.088	0.006	-0.081	-1.15
	8	-0.152	0.006	-0.089	-10.29
	12	-0.226	0.007	-0.106	-17.02
	16	-0.162	0.007	-0.120	-5.97
	20	-0.091	0.007	-0.131	+6.16
Hansen $J = 1822.15$, $\text{dof} = 14$, $\chi^2_{0.05} = 23.68$, $p = 0.000$					

Signs correct for all 16 moments; sectoral ordering reproduced at every horizon. $J = 1822$ (14 d.f.) reflects the high precision of the bank panel—the economically relevant misfit is in the trough depths.

Identification: Capital-Requirement Shocks

Shock series: Drechsel and Miura (2025)

- Narrative + sign-restriction approach
- Captures unanticipated changes in the capital-requirement stance
- Available 2002Q1–2023Q2; estimation sample capped at 2019Q4

Local Projection specification:

$$x_{t+h} - x_{t-1} = \alpha_h + \beta_h \text{shock}_t^{\text{Cap}} + \sum_{p=1}^6 \psi_{h,p} x_{t-p} + \varepsilon_{t+h}$$

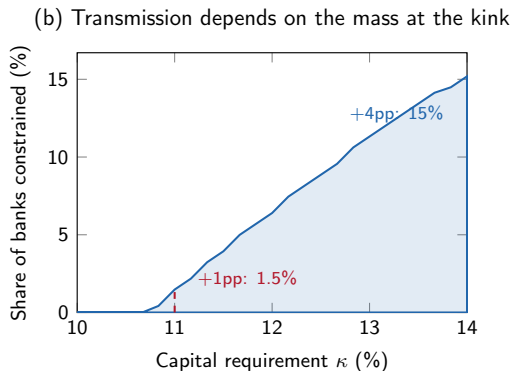
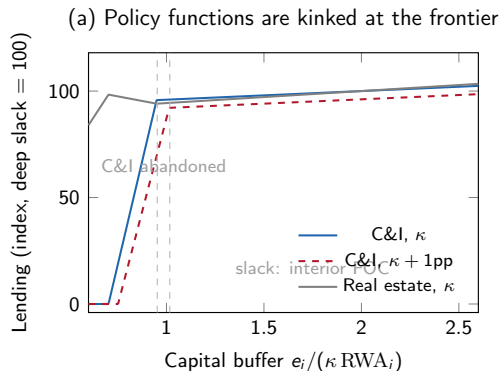
- ▶ **Aggregate:** OLS, Newey–West (lag $h + 6$)
- ▶ **Bank-level:** bank FE, clustered by cert
- ▶ Consumer credit placebo: no response $\Rightarrow$ supply, not demand

The Buffer Gradient: Details

- ▶ Thin-buffer banks cut C&I $1.32\times$ more than thick-buffer banks (90% CI [1.01, 1.62]); $1.41\times$ in real estate, $1.37\times$ pooled [1.13, 1.61] — from a joint cluster-robust covariance across the separate tercile regressions
- ▶ The risk-weight mechanism implies the *same* gradient in both sectors, and the restriction is not rejected: difference $+0.090$ (s.e. 0.186), $t = 0.48$
- ▶ Freezing tercile membership at each bank's pre-sample average slack **reverses** the gradient, to 0.67 in C&I — it is the buffer a bank holds when the shock arrives that matters, not a fixed bank type

Horizon profile, cohort power, and the continuous-interaction specification: see backup.

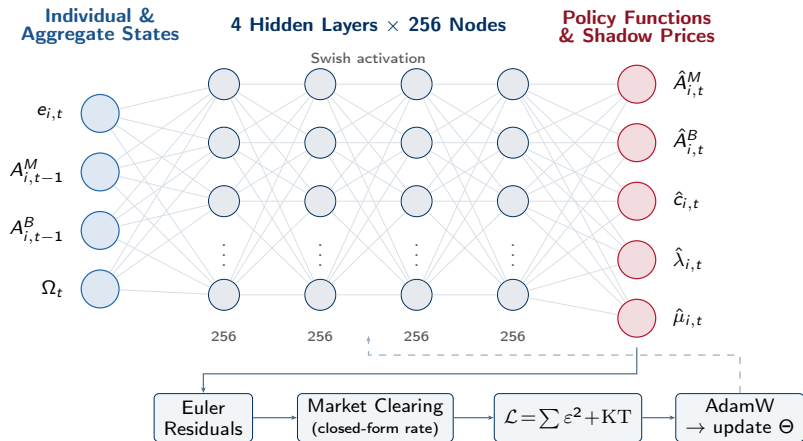
Why a Global Non-Linear Method



Both panels are the *exact* static solution — the problem is quadratic with a linear constraint, so it solves in closed form in each KKT regime.

- ▶ The capital constraint binds occasionally, so the policy function is **kinked**: left of the frontier lending is pinned by equity, right of it the interior FOC holds and lending is flat in equity
- ▶ A +1pp tightening moves the kink *right* — banks that were unconstrained become constrained
- ▶ So the aggregate response is not a fixed elasticity: it depends on the **mass sitting near the kink**. +1pp constrains 1.5% of banks, +4pp constrains 15%

Neural-Network Architecture



Training: $\sim 6h$ on NVIDIA A100 · Cosine annealing schedule · Euler errors $< 10^{-3}$ across state space

[Back](#)

Gross Cost: Numbers

Experiment	Quantity	Value
<i>CEV of +1pp κ tightening (eqs. (??)–(??))</i>		
Total welfare cost	Λ	1.421%
scale channel	Λ_{scale}	1.415%
allocation channel	Λ_{alloc}	0.004%
additivity check	$\Lambda_{scale} + \Lambda_{alloc}$	1.419%
Credit contraction (total / RE / C&I)	ΔL	-3.09% / -3.02% / -3.50%
<i>CEV from the franchise value V (discounted dividends)</i>		
Total welfare cost	Λ_V	2.404%
scale channel	$\Lambda_{V,scale}$	2.392%
allocation channel	$\Lambda_{V,alloc}$	0.026%
additivity check	sum	2.419%
<i>CF1: uniform vs. size-differentiated κ (eq. (??))</i>		
Differentiated rule	Λ	1.236%
Uniform $\tilde{\kappa} = 0.1246$	Λ	1.239%
<i>CF2: flat risk weights $\tilde{\omega} = 0.75$ (eq. (??))</i>		
Baseline sectoral asymmetry	$s_B - s_M$	-2.224
Flat-weight sectoral asymmetry	$s_B - s_M$	-0.000

Credit-market surplus: $\Lambda = 1.4\%$. Franchise value: $\Lambda_V = 2.4\%$. Scale channel accounts for $>99\%$ of the cost. Magnitudes are *directional*: robust in sign and decomposition, sensitive to the solved policy. The cross-sectional gradient is not.

Policy Counterfactuals

CF1: Uniform vs. size-differentiated κ

- ▶ Replace the step function with a uniform $\tilde{\kappa} = 0.1246$ raising the same aggregate capital
- ▶ Gross cost: 2.680% vs. 2.682% $\Rightarrow$ size-tailoring redistributes but barely reduces the aggregate burden

CF2: Risk-weight flattening

- ▶ Set $\tilde{\omega}_B = \tilde{\omega}_M = 0.75$ (average), hold κ fixed
- ▶ Sectoral asymmetry $s_B - s_M$ collapses from -2.22 to 0.00
- ▶ Aggregate credit contraction essentially unchanged

Level of requirements $\Rightarrow$ aggregate cost.
Design (risk weights, size tiers) $\Rightarrow$ incidence.

Backup: Counterfactual Definitions

CF1 — uniform vs. size-differentiated requirements. Replace the statutory step function $\kappa_i(\text{Scale}_i)$ with the single rate $\tilde{\kappa}$ that raises the *same aggregate capital* in the stationary distribution μ^* :

$$\tilde{\kappa} \text{ s.t. } \int \tilde{\kappa} e_i \, d\mu^* = \int \kappa_i(\text{Scale}_i) e_i \, d\mu^*.$$

So the experiment holds the *level* of regulation fixed and varies only its *design*: any difference in Λ is pure incidence.

CF2 — flat risk weights. Set $\omega_M = \omega_B = \tilde{\omega} = 0.75$, removing the risk-weight wedge. The test is whether the two sectors then deleverage proportionally:

$$\frac{\partial \tilde{L}_{t+h}^B}{\partial \varepsilon_t^\kappa} \approx \frac{\partial \tilde{L}_{t+h}^M}{\partial \varepsilon_t^\kappa} \quad \forall h.$$

	Λ	sectoral asymmetry $s_B - s_M$
Differentiated (baseline)	1.236%	-2.224
Uniform $\tilde{\kappa} = 0.1246$	1.239%	—
Flat weights $\tilde{\omega} = 0.75$	—	-0.000

Design redistributes the burden; it barely moves the aggregate. Flattening the weights eliminates the sectoral asymmetry entirely.

Backup: Related Literature

Quasi-experimental credit supply

Aiyar, Calomiris, and Wieladek (2016), Gropp, Guettler, and Saadi (2020), De Jonghe, Dewachter, and Ongena (2020), Jimenez et al. (2017), Conti, Nobili, and Signoretti (2023)

Here: the same identified shock is traced across sectors *and* across the buffer distribution, then mapped into a structural cost.

Deposit franchise and regulatory incidence

Drechsler, Savov, and Schnabl (2017)

The deposit franchise shapes not only monetary transmission but the incidence of *capital* regulation: what disciplines the buffer is the *marginal* funding source.

Quantitative macro-banking

Corbae and Erasmo (2021), Begenau (2020), Elenev, Landvoigt, and Van Nieuwerburgh (2021), Mendicino et al. (2020)

Backup: Reading c and z

$$\phi_{i,j}(\kappa) = (c + z\omega_j) \max(\kappa - \bar{\kappa}_i, 0)$$

- ▶ $z = 0.021$ is **risk-weighted**: a Modigliani–Miller premium, $\approx 10\%$ annualised equity–debt spread
- ▶ $c = 0.148$ is **not** risk-weighted and seven times larger — what a **leverage requirement** imposes. U.S. banks face one; the model's constraint is risk-based only
- ▶ Both are set on the two sectoral aggregates; *both* buffer gradients are untargeted

The leverage-requirement reading is a conjecture about the *source* of the non-risk-weighted term. The data identify its *magnitude* either way.

Backup: SMM Weighting and Algorithm

Weighting matrix

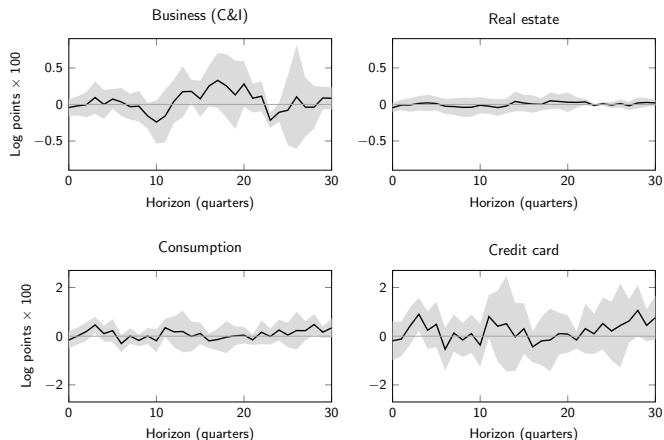
$W = \hat{\Sigma}^{-1}$, block-diagonal, estimated by cluster bootstrap of the bank-level local projections (100 replications)

Algorithm — for each candidate $\theta = (\rho_{\kappa}, \sigma_{\kappa})$

- 1 Re-solve the model globally with the neural network
- 2 Simulate a panel impulse path under the candidate process
- 3 Compute model-implied LP coefficients on the simulated panel
- 4 Evaluate $(\hat{m} - m(\theta))'W(\hat{m} - m(\theta))$

Each evaluation requires a full re-solve, which is what makes the global solution method the binding cost of estimation.

Backup: All Four Loan Categories



C&I contracts most, real estate less, and consumption and credit-card lending do not respond at any horizon. The two unresponsive categories are the demand placebo: they are fully risk-weighted but funded against high interest margins, so a capital tightening need not be absorbed through volume.

Backup: Summary of Empirical Facts

Fact	Finding
1. Sectoral asymmetry	C&I contracts $\sim$ a third more than RE within banks (risk-weight wedge)
2. Buffer gradient	Thin-buffer banks cut $1.32\times$ more; with terciles frozen at pre-sample slack the gradient <i>reverses</i> (0.67)
3. Placebo	Consumption and credit-card lending unresponsive at all horizons
4. Robustness	Results stable to 2023Q2 extension, pandemic controls

Backup: Where the Euler Terms Come From

Law of motion for equity. Every term in the Euler equation is a derivative of this line.

$$e_{i,t+1} = (1 - \delta_e) e_{i,t} + \sum_{j \in \{M, B\}} \left[(r_{j,t} - \varepsilon_{i,t+1}^j) A_{i,t}^j - \frac{\psi_j}{2} (A_{i,t}^j)^2 - \phi_{i,j}(\kappa_{i,t}) A_{i,t}^j \right] \\ - r_{d,t} \bar{d}_{i,C} - r_{W,t} d_{i,t}^W - \frac{\gamma_i}{2} (d_{i,t}^W)^2 - c_{i,t} - \tau_t e_{i,t}$$

Wholesale funding is residual, not chosen: $d_{i,t}^W = A_{i,t}^M + A_{i,t}^B - e_{i,t} - \bar{d}_{i,C}$

$\varepsilon_{i,t+1}^j$: idiosyncratic credit loss, realised after lending, so the net return per dollar is $r_{j,t} - \varepsilon_{i,t+1}^j$

ψ_j : quadratic operating cost — marginal cost $\psi_j A_{i,t}^j$ rises with portfolio size, which pins the long-run sectoral mix

$r_{d,t}$, $r_{W,t}$: retail deposit and wholesale funding rates

δ_e : quarterly equity depreciation (operating costs, franchise decay)

τ_t : systemic insurance levy on equity, funding entrant capital injections

[Back to Problem \(I\)](#)

[Back to Euler \(II\)](#)

Backup: Slack on Funding Capacity

Dep. var: Slack (pp)	(1) Pooled	(2) Between	(3) Within
Funding Capacity	−19.44*** (1.73)	−25.97*** (1.95)	−2.27** (1.05)
Log Assets	−2.80*** (0.09)	−2.70*** (0.10)	−6.27*** (0.16)
Quarter FE / Bank FE	Yes / No	—	Yes / Yes
<i>N</i>	935,234	16,117	935,056

The between estimate (−25.97) is the cross-sectional regularity; the within estimate (−2.27) is small because the funding mix barely moves at quarterly frequency. Size enters separately and negatively throughout, so this is not size in disguise.

Stylized Model: Environment

Static, partial equilibrium. Continuum of risk-neutral banks $i \in [0, 1]$.

- ▶ Two asset classes: mortgages (A_i^M , $\omega_M = 0.50$) and C&I (A_i^B , $\omega_B = 1.00$)
- ▶ Liabilities: core deposits $d_{i,C} \leq \bar{d}_{i,C}$ (cheap, capped) + wholesale $d_{i,W}$ (quadratic friction $\frac{\gamma_i}{2} d_{i,W}^2$)
- ▶ $\gamma_{\text{small}} \gg \gamma_{\text{large}} \rightarrow 0$: small banks face steep wholesale costs
- ▶ Regulatory constraint: $e_i \geq \kappa(\omega_M A_i^M + \omega_B A_i^B)$

Capital buffer:

$$D_i = e_i - \kappa(\omega_M A_i^M + \omega_B A_i^B) \geq 0$$

Banks with high $\gamma_i \Rightarrow$ large endogenous D_i (self-insurance).

First-Order Conditions and the Sectoral Channel

FOC for asset class $j \in \{M, B\}$:

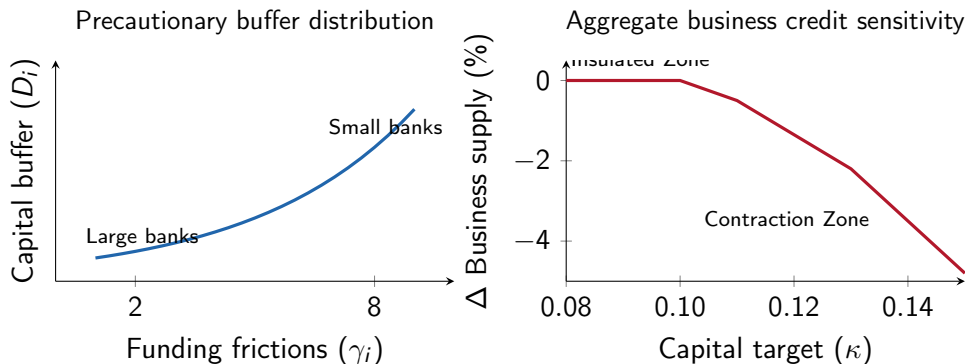
$$r_j - r_W - \psi_j A_i^j - \gamma_i (A_i^M + A_i^B - e_i - d_{i,C}) = \lambda_i \kappa \omega_j$$

When $\lambda_i > 0$ (constraint binds):

- ▶ **Asset Substitution Margin:** $|\partial A_i^B / \partial \kappa| > |\partial A_i^M / \partial \kappa|$ because $\omega_B > \omega_M$
- ▶ **Scale Deleveraging Multiplier:** $\partial RWA / \partial A^B = \omega_B = 1 > 0.5 = \omega_M \Rightarrow$ cutting C&I frees 2× the regulatory capital per dollar

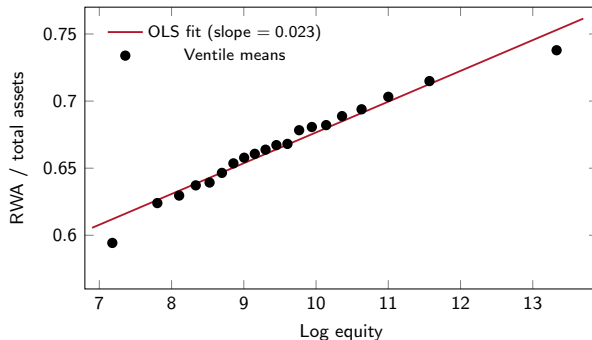
Cross-section: for low- γ banks, λ_i stays low $\Rightarrow$ small response. For high- γ banks whose buffer is breached, λ_i spikes $\Rightarrow$ acute, non-linear contraction in C&I.

The Insulated Zone



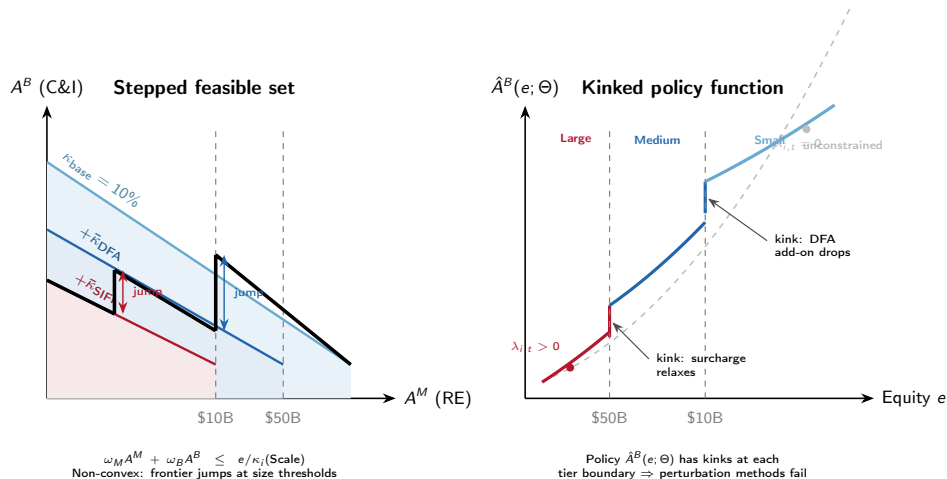
When shocks are small-to-moderate, well-buffered banks absorb them internally. Only shocks large enough to breach the buffer trigger non-linear credit contractions.

The Regulatory Wedge



RWA/total-assets ratio rises with bank equity $\Rightarrow$ larger banks hold higher average risk weights; portfolio composition drives the regulatory position of small banks.

Why Neural Networks? The Non-Convexity Challenge



Size-tiered surcharges create a stepped feasible set (left) and kinks in optimal policy (right). Standard perturbation around the steady state cannot capture these discontinuities—motivating a global neural-network solution.

Calibration

Parameter	Description	Value	Source
β	Discount factor	0.99	Standard
δ_e	Equity depreciation	0.001	Stationarity
ω_M, ω_B	Risk weights	0.50, 1.00	Basel III
κ_{base}	PCA minimum	10%	Regulatory
$\bar{\kappa}_{\text{DFA}}$	DFA add-on	1.5%	Enhanced prudential
$\bar{\kappa}_{\text{SIFI}}$	SIFI add-on	2.5%	G-SIB surcharge
γ_{small}	Wholesale friction	0.077	Funding structure
ψ_M	RE adjustment cost	0.007	RE share target
ψ_B	C&I adjustment cost	0.05	Preset
η_M, η_B	Demand slopes	external	Lit. semi-elasticities
$\sigma_{\varepsilon, M}$	Idiosync. shock (RE)	0.001	Charge-off data
$\sigma_{\varepsilon, B}$	Idiosync. shock (C&I)	0.004	Charge-off data

Consumption-equivalent variation (CEV):

$$V^{\kappa}(\mu^*; \kappa^* + 0.01) = V^*((1 - \Lambda)\mu^*; \kappa^*)$$

$\Lambda > 0$ = fraction of steady-state surplus households would forgo to avoid the tightening.

Decomposition:

$$\Lambda = \underbrace{\Lambda^{\text{alloc}}}_{\text{sectoral distortion}} + \underbrace{\Lambda^{\text{scale}}}_{\text{credit contraction}}$$

- ▶ Two metrics: credit-market surplus and franchise value (discounted dividends)
- ▶ **Caveat:** gross cost—no stability benefits, partial equilibrium

Backup: Robustness to Sample Extension

Re-estimating small-cohort LPs on the full shock series (2002Q1–2023Q2):

Treatment	Mean C&I response	Mean RE response
Baseline (2002–2019)	−0.145%	−0.105%
Pandemic dummy $\times$ shock	−0.142%	−0.106%
Exclude pandemic shocks	−0.142%	−0.106%

Target moments are essentially unchanged.

[Sensitivity to \$\psi_B\$](#)

[Back](#)

Backup: Sensitivity to ψ_B

Constrained-regime sectoral asymmetry $s_B - s_M$ across the admissible range of $\psi_B \in (0.014, 0.142)$:

ψ_B	$s_B - s_M$
0.02	-3.5
0.05 (baseline)	-2.2
0.10	-0.8
Flat weights (any ψ_B)	0.0

The asymmetry is a risk-weight phenomenon; ψ_B governs only its amplitude.

Backup: Precision of the Buffer Gradient

Terciles are estimated as separate local projections, so the *difference* needs a joint covariance: a bank contributes scores to several tercile blocks over its life, and the twenty horizons use overlapping windows.

	Thin	Thick	Ratio (90% CI)
C&I	−0.1718	−0.1307	1.32 [1.01, 1.62]
Real estate	−0.1354	−0.0964	1.41 [1.14, 1.67]
Pooled	—	—	1.37 [1.13, 1.61]

The cross-sector restriction is not rejected. The risk-weight mechanism implies the same gradient in both sectors: measured difference +0.090 (s.e. 0.186), $t = 0.48$, across 9,801 banks.

[Back](#)

Backup: What the Solved Policy Does and Does Not Pin Down

The model is solved by training a policy network. What that choice affects is not what an earlier version of this slide claimed.

Stable across solved policies. The welfare cost. With the cost of regulatory capital active, Λ lies between 1.16% and 1.42% across solved lending from $0.21\times$ to $1.05\times$ the calibration target — it tracks the *proportional* contraction ($\approx -3\%$), not the level.

Not stable. The impulse-response magnitudes, and the buffer gradient when it is recomputed *from* a solved policy rather than from the static problem.

So the gradient we report is the exact one. The static problem is quadratic with a linear constraint, so it solves in closed form in each KKT regime: C&I 1.32 against 1.32 in the data. The trained policy returns 1.02–1.06 on the same calibration — it smooths the kink at the frontier, where its residuals are largest — and neither added capacity nor a four-fold longer schedule closes the gap.

What collapses Λ to 0.1% is not a low solved level but *removing* $\phi_{i,j}$: without it total credit moves -0.25% rather than -3.1% . That is a different model, not a numerical sensitivity — and it is the sharpest evidence that $\phi_{i,j}$, not the capital constraint, generates the welfare cost.

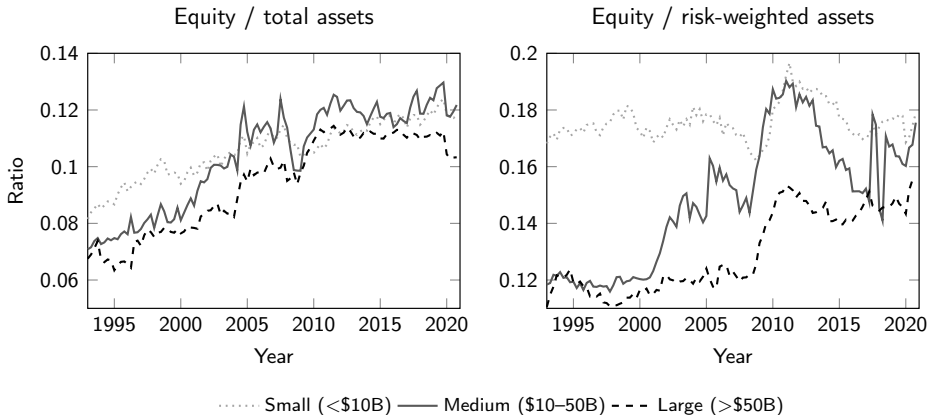
Backup: Sensitivity to the Demand Calibration

η_j enters neither the bank's problem nor the reduced-form estimates, so the lending responses are *exactly* invariant to it. The welfare *level* is not, because the surplus integral is taken under the demand curve.

S_M	S_B	Λ
32.5 (implemented)	14.8	1.42%
16.3	7.4	0.95%
2.0 (cited targets)	4.0	0.27%

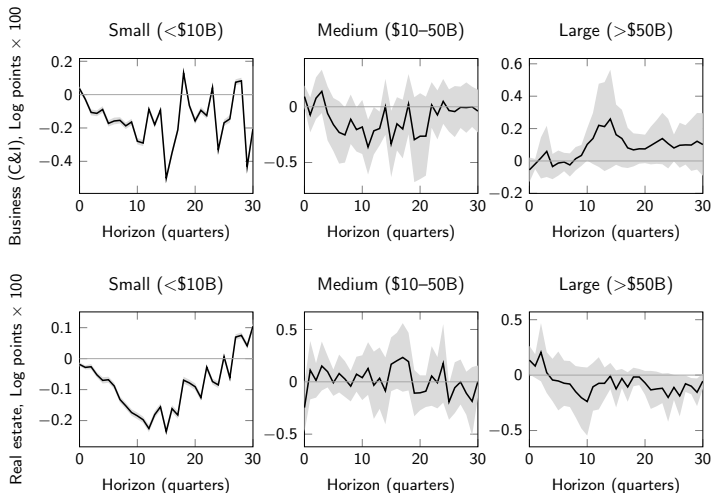
The externally cited estimates measure objects adjacent to a loan-demand schedule — a house-price elasticity, and a loan-*granting* probability against the policy rate — rather than the schedule itself.

Backup: Leverage vs. Regulatory Capital Ratios



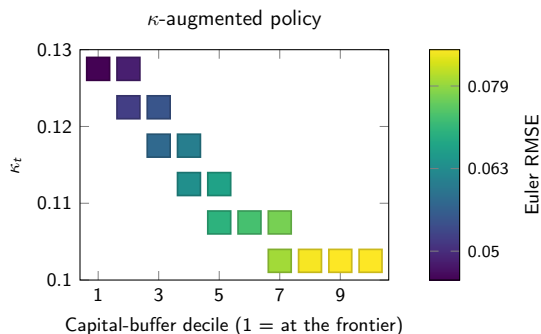
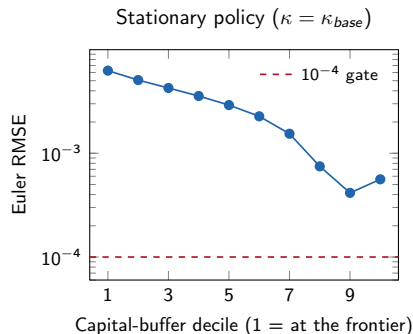
Small banks appear more capitalised on an unweighted basis, but the advantage compresses under risk-weighting.

Backup: Size Cohort Impulse Responses



Small-bank panel precisely estimated; large-bank cohort (≈ 40 institutions) is not. The buffer distribution, not size per se, drives the cross-section.

Backup: Solution Accuracy



All deciles below 10^{-2} . Frontier spike reflects the kink in policy functions at the occasionally binding constraint.

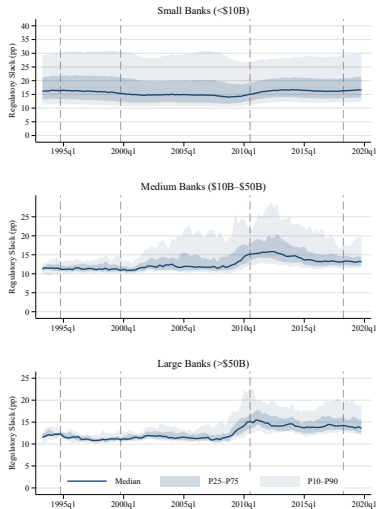
Train vs. test

Ergodic coverage

Back

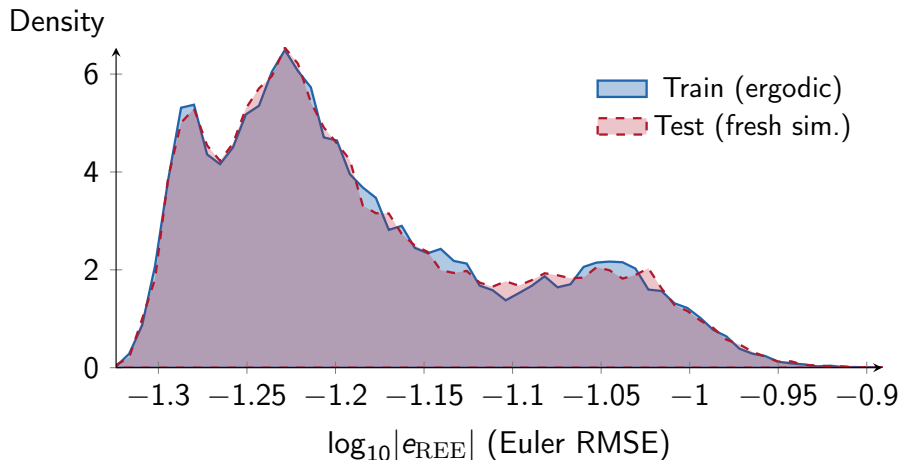
Backup: Slack Distribution by Size, 1993Q2–2019Q4

Distribution of Regulatory Slack by Bank Size



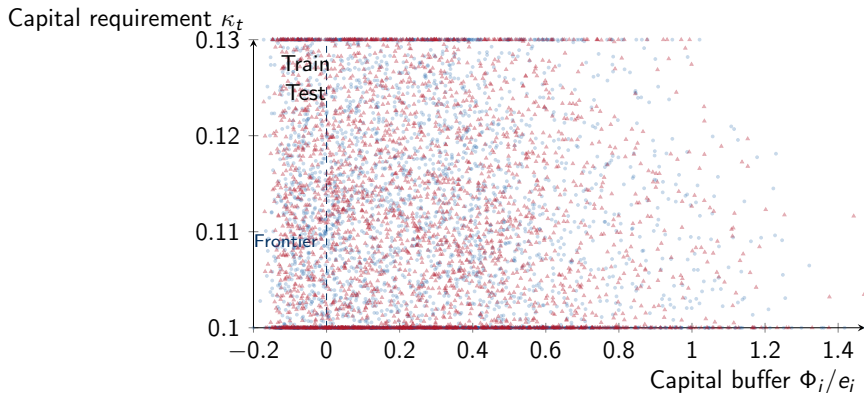
Shaded bands: P10–P90 (light) and P25–P75 (dark). Solid line: median.
Dashed vertical lines mark Riegle-Neal (1994Q4), GLBA (1999Q4),
Dodd-Frank (2010Q3, highlighted), and EGRRCFA (2018Q2).
Sample: quarterly bank panel, 1993Q2–2019Q4. Source: US Call Reports.

Backup: Euler Errors, Training vs. Test



Distribution of $\log_{10}|e_{\text{REE}}|$ on the training panel (seed 0) and an independent test panel (seed 42), 20,000 banks each after 200 periods of burn-in. Median residual -1.21 in *both* samples and no fattening of the test tail: the network generalises to unseen states rather than memorising the training trajectory.

Backup: Ergodic-Set Coverage



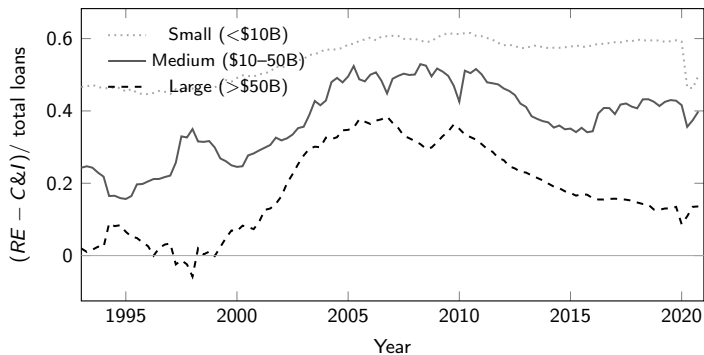
Joint distribution of the capital buffer Φ_i/e_i and the requirement κ_t , training against test simulation. The test cloud lies entirely inside the training cloud's convex hull, so the policy is evaluated only where it was solved — extrapolation beyond the training support is the most common failure mode of neural-network approximations.

Backup: Magnitudes in Context

Study	Design / setting	Effect of +1pp required capital
Aiyar, Calomiris, and Wieladek (2016)	UK bank-level requirement changes, 1998–2007	Lending growth 6.5–7.2pp lower
Mésonnier and Monk (2015)	EBA 2011–12 capital exercise, euro area	Annualised loan growth 1.2–1.6pp lower
De Jonghe, Dewachter, and Ongena (2020)	Pillar-2 decisions, Belgian credit register, 2013–15	Corporate credit growth ≈ 0.2 – 0.4 pp per quarter lower
Conti, Nobili, and Signoretti (2023)	Narrative BVAR, Italy (EBA/CA episodes)	Loan supply -1.5% after 2 years per +40bp Tier 1 ($\approx -3.8\%$ per pp, linearised)
This paper (model)	SMM-estimated policy, permanent +1pp κ	C&I -3.5% , RE -3.0% , total credit -3.1% (stationary comparison)
This paper (LPs)	Within-bank LPs scaled through the SMM mapping (one shock unit $\mapsto \hat{\sigma}_{\kappa} = 4.0$ bp of κ)	Trough responses $\approx -8.8\%$ (C&I, $h=16$) and $\approx -5.7\%$ (RE, $h=12$); out-of-sample linear extrapolation

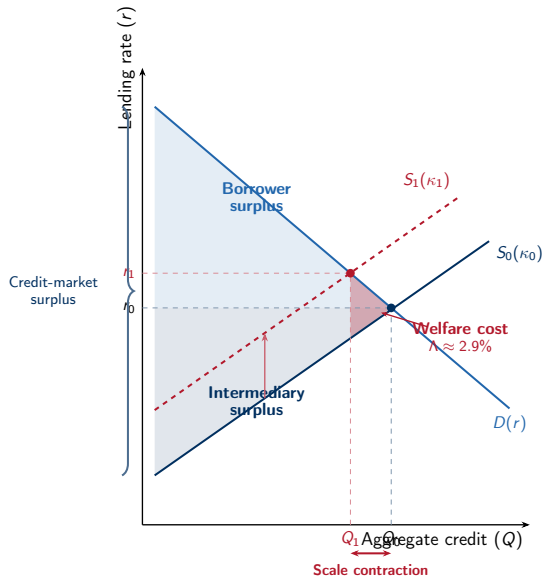
Model's -3.5% C&I / -3.0% RE / -3.1% total per +1pp sits squarely inside the quasi-experimental range. These magnitudes are conditional on the solved policy; the cross-sectional gradient is not.

Backup: Asset Gap



Difference between RE and C&I lending as a share of total loans, by size cohort, 1993–2020.

Welfare: Credit-Market Surplus



References

-  Aiyar, Shekhar, Charles W Calomiris, and Tomasz Wieladek (2016). “How does credit supply respond to monetary policy and bank minimum capital requirements?” In: *European Economic Review* 82, pp. 142–165.
-  Begenau, Juliane (2020). “Capital requirements, risk choice, and liquidity provision in a business-cycle model”. In: *Journal of Financial Economics* 136.2, pp. 355–378.
-  Conti, Antonio M, Andrea Nobili, and Federico M Signoretti (2023). “Bank capital requirement shocks: A narrative perspective”. In: *European Economic Review* 151, p. 104254.
-  Corbae, Dean and Pablo D Erasmo (2021). “Capital Buffers in a Quantitative Model of Banking Industry Dynamics”. In: *Econometrica* 89.6, pp. 2975–3023. DOI: [10.3982/ecta16930](https://doi.org/10.3982/ecta16930).
-  De Jonghe, Olivier, Hans Dewachter, and Steven Ongena (2020). “Bank capital (requirements) and credit supply: Evidence from pillar 2 decisions”. In: *Journal of Corporate Finance* 60, p. 101518.